

RAMCO INSTITUTE OF TECHNOLOGY

**DEPARTMENT OF ELECTRICAL AND ELECTRONICS
ENGINEERING**

COURSE OUTCOMES

Regulation R-2021

Course Code & Title : HS 3151 Professional English - I	
Year / Semester : I Year/I Semester	
Course Index : C101	
C101.1	Communicate clearly both in written and oral forms using appropriate vocabulary and comprehend written text to make inferences
C101.2	Speak persuasively in academic/work contexts and write biographical details and technical documents cohesively, coherently and flawless using appropriate words
C101.3	Read, write and speak effectively in a variety of professional and social settings
C101.4	Comprehend descriptive, narrative, expository and interpretive texts and write using creative, critical, analytical and evaluative methods
C101.5	Understand and respond to different spoken and written discourses/excerpts in different accents and write different genres of text adopting various writing strategies
C101.6	Detect paralinguistic cues such as postures, gestures, facial expressions and eye contact
Course Code & Title : MA3151 Matrices and Calculus	
Year / Semester : I Year/I Semester	
Course Index : C102	
C102.1	Use the matrix algebra methods for solving practical problems.
C102.2	Apply differential calculus tools in solving various application problems.
C102.3	Use differential calculus ideas on several variable functions.
C102.4	Apply different methods of integration in solving practical problems.
C102.5	Apply multiple integral ideas in solving areas, volumes and other practical problems.
Course Code & Title : PH3151 Engineering Physics	
Year / Semester : I Year/I Semester	
Course Index : C103	
C103.1	Understand the importance of mechanics.
C103.2	Express their knowledge in electromagnetic waves.
C103.3	Demonstrate a strong foundational knowledge in oscillations, optics and lasers.
C103.4	Understand the importance of quantum physics.
C103.5	Comprehend and apply quantum mechanical principles towards the formation of energy bands.
C103.6	Build a simulation models to calculate the energy values of particle in a 1D, 2D & 3D boxes using computer programming
Course Code & Title : CY 3151 Engineering Chemistry	
Year / Semester : I Year/I Semester	
Course Index : C104	
C104.1	Illustrate the importance of water quality parameters, water treatment methods and boiler troubles.
C104.2	Explain the types of nanomaterials, synthesis and its applications.
C104.3	Apply the use of phase rule in metallurgy, basics of composites & its applications.
C104.4	Articulate the fuel types, synthesis and its combustion characteristics
C104.5	Examine the working principle of alternate energy resources and energy storage devices.
C104.6	Analyze the water quality parameters of water samples collected in and around the native area

Course Code & Title : GE3151 Problem Solving and Python Programming	
Year / Semester : I Year/I Semester	
Course Index : C105	
C105.1	Develop algorithmic solutions to simple computational problems
C105.2	Demonstrate simple Python Programs
C105.3	Apply control structures, functions and string for solving problems
C105.4	Implement compound data using Python list, tuples, dictionaries etc
C105.5	Illustrate read and write data from/to files in python programs
C105.6	Build an application by applying varying python concepts
Course Code & Title : GE3171 Problem Solving and Python ProgramminG Laboratory	
Year / Semester : I Year/I Semester	
Course Index : C106	
C106.1	Develop diagrammatic solutions to simple computational problems
C106.2	Execute simple python programs using statements, operators and expresiions
C106.3	Implement various computing strategies for python-based solutins to real world problems
C106.4	Apply compound data using python lists, tuples, sets and dictionaries to solve real life problems
C106.5	Implement read and write data from/to files in python programs
Course Code & Title : BS3171 Physics and Chemistry Laboratory	
Year / Semester : I Year/I Semester	
Course Index : C107	
C107.1	Understand the functioning of various physics laboratory equipment.
C107.2	Use graphical models to analyze laboratory data.
C107.3	Use mathematical models as a medium for quantitative reasoning and describing physical reality.
C107.4	Analyze the quality of the water samples like hardness, alkalinity, chloride and dissolved oxygen content
C107.5	Quantitatively examine the impurities present in solutions by electro-analytical techniques such as pH, Conductometry.
Course Code & Title : HS3251 Professional English - II	
Year / Semester : I Year/II Semester	
Course Index : C108	
C108.1	Communicate effectively in written and spoken formats in professional settings using suitable contextual word and compare and contrast products and ideas in technical texts
C108.2	Identify cause and effects in events, describe industrial processes through technical texts and respond to questions using appropriate vocabulary
C108.3	Analyze problems in order to arrive at feasible solutions and communicate them orally and in the written format
C108.4	Report events and the processess of technical and industrial nature in written and oral forms
C108.5	Present opinions in a planned and logical manner and draft effective resumes in context of job search
C108.6	Respond to various non-verbal signals such as stances,motions,kinesics and proxemics

Course Code & Title : MA3251 Statistics and Numerical Methods	
Year / Semester : I Year/II Semester	
Course Index : C109	
C109.1	Apply the concept of testing of hypothesis for small and large samples in real life problems.
C109.2	Apply the basic concepts of classifications of design of experiments in the field of agriculture.
C109.3	Appreciate the numerical techniques of interpolation in various intervals and apply the numerical techniques of differentiation and integration for engineering problems.
C109.4	Understand the knowledge of various techniques and methods for solving first and second order ordinary differential equations.
C109.5	Solve the partial and ordinary differential equations with initial and boundary conditions by using certain techniques with engineering applications
Course Code & Title : PH3202 Physics for Electrical Engineering	
Year / Semester : I Year/II Semester	
Course Index : C110	
C110.1	Elaborate basics of dielectric materials and insulation.
C110.2	Acquire knowledge on the electrical and magnetic properties of materials and their applications
C110.3	Understand clearly of semiconductor physics and functioning of semiconductor devices
C110.4	Understand the optical properties of materials and working principles of various optical devices
C110.5	Appreciate the importance of nanotechnology and nanodevices.
C110.6	Build a simulation model/programme to predict the behavior of electrons
Course Code & Title : BE3255 Basic Civil and Mechanical Engineering	
Year / Semester : I Year/II Semester	
Course Index : C111	
C111.1	Understanding profession of Civil and Mechanical engineering
C111.2	Summarise the planning of building, infrastructure and working of Machineries.
C111.3	Apply the knowledge gained in respective discipline
C111.4	Illustrate the ideas of Civil and Mechanical Engineering applications.
C111.5	Appraise the material, Structures, machines and energy.
C111.6	Display the operation of two- and four-stroke engines
Course Code & Title : GE3251 Engineering Graphics	
Year / Semester : I Year/II Semester	
Course Index : C112	
C112.1	Construct the conic sectional curves, cycloidal and involute curves.
C112.2	Solve practical problems involving projection of lines.
C112.3	Construct the projection of simple solids and free hand sketches of multiple views from pictorial views of objects.
C112.4	Sketch the Section of Solids and Development of lateral Surfaces of solids.
C112.5	Illustrate the isometric and perspective projections of simple solids.

Course Code & Title : EE3251 Electric Circuit Analysis Year / Semester : I Year/II Semester Course Index : C113	
C113.1	Analyse the DC and AC circuits using ohm's law, Kirchhoff's laws, mesh current and node voltage method.
C113.2	Solve DC and AC circuits using network reduction techniques and theorems
C113.3	Analyse the transient response of first order and second order systems for source free, step and sinusoidal inputs
C113.4	Explain the frequency response of series and parallel resonance circuits and behavior of magnetically coupled circuits.
C113.5	Compute line currents, line voltages, phase currents, phase voltages, power and power factor in balanced and unbalanced three-phase circuits.
C113.6	Build a DC circuit hardware set up for implementing circuit laws and network theorems
C113.7	Present the hardware set up as a team, interact with team members in a professional manner, respecting differences, to ensure a collaborative project environment.
Course Code & Title : GE3271 Engineering Practices Laboratory Year / Semester : I Year/II Semester Course Index : C114	
C114.1	Draw pipe line plan; lay and connect various pipe fittings used in common household plumbing work; Saw; plan; make joints in wood materials used in common household wood work.
C114.2	Construct various electrical joints in common household electrical wire work.
C114.3	Perform welding various joints in steel plates using arc welding work; Machine various simple processes like turning, drilling, tapping in parts; Assemble simple mechanical assembly of common household equipments; Make a tray out of metal sheet using sheet metal work.
C114.4	Perform soldering and test simple electronic circuits; Assemble and test simple electronic components on PCB.
C114.5	Demonstrate assemble and dismantle of LED TV and computer/ laptop components.
C114.6	Model a circuit design using passive and active component in virtual lab.
Course Code & Title : EE3271 Electric Circuits Laboratory Year / Semester : I Year/II Semester Course Index : C115	
C115.1	Solve the series and parallel electrical circuit using fundamental laws.
C115.2	Experiment the various electrical theorems for the given DC/AC circuits.
C115.3	Analyze the transient behavior of the given RL/RC/RLC circuit.
C115.4	Show the frequency response of the series and parallel RLC circuit.
C115.5	Examine the Power and Power factor calculations for the three phase star and delta circuits.

Course Code & Title : MA3303 Probability and Complex Functions Year / Semester : II Year/III Semester Course Index : C201	
C201.1	Understand the fundamental knowledge of the concepts of probability and have knowledge of standard distributions which can describe real life phenomenon.
C201.2	Understand the basic concepts of one and two dimensional random variables and apply in engineering applications.
C201.3	Develop an understanding of the standard techniques of complex variable theory in particular analytic function and its mapping property.
C201.4	Familiarize the students with complex integration techniques and contour integration techniques which can be used in real integrals.
C201.5	Acquaint the students with Differential Equations which are significantly used in engineering problems.
Course Code & Title : EE3301 Electromagnetic Fields Year / Semester : II Year/III Semester Course Index : C202	
C202.1	Interpret the effect of static electric fields in electrical engineering applications.
C202.2	Evaluate the electric field intensity and electric potential for electrical engineering applications.
C202.3	Evaluate the magnetic field intensity and magnetic potential for electrical engineering applications.
C202.4	Interpret the model of electromagnetic fields using Maxwell's equation.
C202.5	Interpret the phenomena of wave propagation in different medium.
Course Code & Title : EE3302 Digital Logic Circuits Year / Semester : II Year/III Semester Course Index : C203	
C203.1	Choose the relevant digital logic family and number system in Integrated circuits
C203.2	Interpret the process for the desired functioning of combinational circuits
C203.3	Interpret the performance of synchronous sequential circuits
C203.4	Interpret the performance of Asynchronous sequential circuits and Programmable Logic Devices
C203.5	Interpret the VHDL Code for proper functioning of combinational and sequential circuits
Course Code & Title : EC3301 Electron Devices and Circuits Year / Semester : II Year/III Semester Course Index : C204	
C204.1	Interpret the functioning of the PN junction devices.
C204.2	Interpret the functioning of transistors and thyristors for various applications.
C204.3	Interpret the functioning of BJT and FET based amplifiers
C204.4	Interpret the functioning of multistage and differential amplifiers
C204.5	Interpret the functioning feedback amplifiers and oscillators for the electronics applications.

Course Code & Title : EE3303 Electrical Machines - I Year / Semester : II Year/III Semester Course Index : C205	
C205.1	Interpret the electro-mechanical energy conversion process in the magnetic systems.
C205.2	Interpret the performance of the DC Generators.
C205.3	Evaluate the performance of the DC Motors.
C205.4	Interpret the performance of the Single Phase Transformers.
C205.5	Evaluate the performance of the Three Phase Transformers and Auto-Transformers.
Course Code & Title : CS3353 C Programming and Data Structures Year / Semester : II Year/III Semester Course Index : C206	
C206.1	Develop C programs for any real world/technical application.
C206.2	Apply advanced features of C in solving problems.
C206.3	Write functions to implement linear and non-linear data structure operations.
C206.4	Suggest and use appropriate linear/non-linear data structure operations for solving a given problem.
C206.5	Appropriately use sort and search algorithms for a given application.
C206.6	Apply appropriate hash functions that result in a collision free scenario for data storage and retrieval.
Course Code & Title : EC3311 Electronic Devices and Circuits Laboratory Year / Semester : II Year/III Semester Course Index : C207	
C207.1	Interpret the functioning of the PN junction devices.
C207.2	Interpret the functioning of transistors and thyristors for various applications.
C207.3	Interpret the functioning of BJT and FET based amplifiers
C207.4	Interpret the functioning of multistage and differential amplifiers
C207.5	Interpret the functioning feedback amplifiers and oscillators for the electronics applications.
Course Code & Title : EE3311 Electrical Machines Laboratory- I Year / Semester : II Year/III Semester Course Index : C208	
C208.1	Interpret the electro-mechanical energy conversion process in the magnetic systems.
C208.2	Interpret the performance of the DC Generators.
C208.3	Evaluate the performance of the DC Motors.
C208.4	Interpret the performance of the Single Phase Transformers.
C208.5	Evaluate the performance of the Three Phase Transformers and Auto-Transformers.

Course Code & Title : CS3362 C Programming and Data Structures Laboratory Year / Semester : II Year/III Semester Course Index : C209	
C209.1	Use different constructs of C and develop applications.
C209.2	Write functions to implement linear and non-linear data structure operations
C209.3	Suggest and use the appropriate linear / non-linear data structure operations for a given problem
C209.4	Apply appropriate hash functions that result in a collision free scenario for data storage and Retrieval
C209.5	Implement Sorting and searching algorithms for a given application
Course Code & Title : GE3361 Professional Development Year / Semester : II Year/III Semester Course Index : C210	
C210.1	Use MS Word to create quality documents, by structuring and organizing content for their day to day technical and academic requirements.
C210.2	Use MS EXCEL to perform data operations and analytics, record, retrieve data as per requirements and visualize data for ease of understanding.
C210.3	Use MS PowerPoint to create high quality academic presentations by including common tables, charts, graphs, interlinking other elements, and using media objects.
Course Code & Title : GE3451 Environmental Sciences and Sustainability Year / Semester : II Year/IV Semester Course Index : C211	
C211.1	Illustrate the importance of environment, biodiversity and its impact on human life.
C211.2	Correlate the causes, effect and control measures of various types of pollution and its case studies.
C211.3	Educate the availability of renewable energy resources and scientific concepts/principles behind them.
C211.4	Articulate the concept of sustainability and management.
C211.5	Gain knowledge on sustainability practices and socio-economical change
C211.6	Analyze the integrated themes of biodiversity, pollution control, waste management, energy resources, and sustainable approaches.
Course Code & Title : EE3401 Transmission and Distribution Year / Semester : II Year/IV Semester Course Index : C212	
C212.1	Compute the transmission line parameters
C212.2	Evaluate the performance of the transmission lines
C212.3	Select relevant towers and insulators for constructing overhead lines.
C212.4	Evaluate the performance of the underground cables
C212.5	Choose the type of distribution system for different applications
C212.6	Evaluate the real and reactive power of transmission lines using power world simulator for given bus system

Course Code & Title : EE3402 Linear Integrated Circuits	
Year / Semester : II Year/IV Semester	
Course Index : C213	
C213.1	Interpret the monolithic IC fabrication of discrete electronic components and PV cells.
C213.2	Interpret the characteristics of the operational amplifiers
C213.3	Evaluate the performance of the operational amplifiers.
C213.4	Evaluate the performance of special function ICs.
C213.5	Evaluate the performance of signal generators and regulators using special function ICs
Course Code & Title : EE3403 Measurements and Instrumentation	
Year / Semester : II Year/IV Semester	
Course Index : C214	
C214.1	Evaluate the characteristics of measuring instrument
C214.2	Interpret the working of various types of instruments for measuring parameters in Electrical system
C214.3	Interpret the working of various AC and DC Bridge circuit and Instrumentation amplifier
C214.4	Interpret the working of various types of sensors and transducers for measuring Non Electrical parameters
C214.5	Interpret the working of various types of instruments used in digital Instrumentation
Course Code & Title : EE3405 Electrical Machines -II	
Year / Semester : II Year/IV Semester	
Course Index : C215	
C215.1	Evaluate the performance of the synchronous generator.
C215.2	Evaluate the performance of the synchronous motor.
C215.3	Evaluate the performance of the three phase induction motor.
C215.4	Suggest the relevant control operations of the three phase induction motor.
C215.5	Evaluate the performance of the single phase induction motor and special machines.
Course Code & Title : EE3411 Electrical Machines Laboratory - II	
Year / Semester : II Year/IV Semester	
Course Index : C216	
C216.1	Evaluate the performance of the synchronous generator.
C216.2	Evaluate the performance of the synchronous motor.
C216.3	Evaluate the performance of the three phase induction motor.
C216.4	Suggest the relevant control operations of the three phase induction motor.
C216.5	Evaluate the performance of the single phase induction motor and special machines.
Course Code & Title : EE3412 Linear and Digital Circuits Laboratory	
Year / Semester : II Year/IV Semester	
Course Index : C217	
C217.1	Construct combinational logic circuits using logic gates and IC's.
C217.2	Build the Counters and Shift registers using flip flops and logic gates.
C217.3	Sketch the output of waveform generator circuits using 555 timer IC.
C217.4	Demonstrate the applications of Op-Amp such as inverting amplifier, non-inverting amplifier, adder, comparator, integrator and differentiator.
C217.5	Examine the voltage to frequency characteristics and the variability voltage regulator performance using special function ICs.

Course Code & Title : EE3413 Microprocessor and Microcontroller laboratory	
Year / Semester : II Year/IV Semester	
Course Index : C218	
C218.1	Interpret the 8085 microprocessor architecture used in various application
C218.2	Evolve the assembly language programs for 8085 microprocessor
C218.3	Evaluate the performance of the peripherals interfaced with 8085 microprocessor
C218.4	Interpret the 8051 microcontroller architecture used in various application
C218.5	Develop the assembly language programs for PIC Controller
Course Code & Title : EE3501 Power System Analysis	
Year / Semester : III Year/V Semester	
Course Index : C301	
C301.1	Interpret the functioning of power system components
C301.2	Evaluate the load flow of power system.
C301.3	Interpret the parameters related to symmetrical faults in power system
C301.4	Interpret the parameters related to unsymmetrical faults in power system.
C301.5	Evaluate the stability of power system
Course Code & Title : EE3591 Power Electronics	
Year / Semester : III Year/V Semester	
Course Index : C302	
C302.1	Interpret the performance of switched mode power supplies.
C302.2	Evaluate the performance of inverters.
C302.3	Interpret the operation of uncontrolled rectifiers.
C302.4	Evaluate the performance of controlled rectifiers.
C302.5	Interpret the operation of AC phase controllers.
Course Code & Title : EE3503 Control Systems	
Year / Semester : III Year/V Semester	
Course Index : C303	
C303.1	Evolve the mathematical models of the physical control system
C303.2	Evaluate the stability of closed loop control systems using time response characteristics
C303.3	Evaluate the stability of closed loop control system using Bode, Polar and Nyquist plot
C303.4	Determine the internal state of the control system using state variable analysis
C303.5	Design the compensator and controller for stable control system
Course Code & Title : EE3032 Energy Storage Systems	
Year / Semester : III Year/V Semester	
Course Index : C304	
C304.1	Compare the working of the different types of Energy meter
C304.2	Interpret the performance of Thermal Energy storage systems
C304.3	Interpret the performance of Electrical Energy Storage systems
C304.4	Interpret the performance of different types of fuel cell Technologies
C304.5	Interpret the performance of various types of alternate energy storage technologies

Course Code & Title : EE3037 Power System Transients Year / Semester : III Year/V Semester Course Index : C305	
C305.1	Evaluate the impact of transients on power system component models.
C305.2	Evaluate the effect of the switching transients on power system equipment.
C305.3	Evaluate the transmission and distribution systems protection with respect to lightning strikes.
C305.4	Evaluate the behaviour of travelling waves and their impact on transmission lines using relevant software.
C305.5	Evaluate the impact of different types of transients in power system using relevant software.
Course Code & Title : CEI331 PLC Programming Year / Semester : III Year/V Semester Course Index : C306	
C306.1	Evaluate the PLC program for its proper functioning.
C306.2	Evaluate the communication of the PLC with other devices.
C306.3	Network the PLC for effective SCADA communication.
C306.4	Implement the HART and Field bus communication protocol.
C306.5	Program the PLCs for the various industry applications.
Course Code & Title : MX3084 Disaster Risk Reduction and Management Year / Semester : III Year/V Semester Course Index : C307	
C307.1	Explain the types, causes and impacts of hazards, vulnerability and disaster risks on environment and society
C307.2	Demonstrate the framework and resources available for disaster risk reduction.
C307.3	Explain various methods of management against risk and disaster as per government acts and policies.
C307.4	Interpret the functioning of tools and technology for disaster management
C307.5	Interpret the impact and actions of hazards and vulnerability profile for different case studies
Course Code & Title : EE3511 Power Electronics Laboratory Year / Semester : III Year/V Semester Course Index : C308	
C308.1	Interpret the performance of switched mode power supplies.
C308.2	Evaluate the performance of inverters.
C308.3	Interpret the operation of uncontrolled rectifiers.
C308.4	Evaluate the performance of controlled rectifiers.
C308.5	Interpret the operation of AC phase controllers.
Course Code & Title : EE3512 Control and Instrumentation Laboratory Year / Semester : III Year/V Semester Course Index : C309	
C309.1	Evolve the mathematical models of the physical control system
C309.2	Evaluate the stability of closed loop control systems using time response characteristics
C309.3	Evaluate the stability of closed loop control system using Bode, Polar and Nyquist plot
C309.4	Determine the internal state of the control system using state variable analysis
C309.5	Design the compensator and controller for stable control system

Course Code & Title : EE3601 Protection and Switchgear	
Year / Semester : III Year/VI Semester	
Course Index : C310	
C310.1	Use the protection schemes judiciously and safely.
C310.2	Evaluate the performance of the electromagnetic relays
C310.3	Interpret the suitable protection system for the apparatus protection.
C310.4	Evaluate the performance of the Static relays and Numerical protection.
C310.5	Choose the type of circuit breaker for different applications
Course Code & Title : EE3602 Power System Operation and Control	
Year / Semester : III Year/VI Semester	
Course Index : C311	
C311.1	Interpret the various power system parameters
C311.2	Evaluate the real power-frequency control mechanisms for stable power systems
C311.3	Evaluate the reactive power-frequency control mechanisms for stable power systems.
C311.4	Evaluate the power system parameters for economic electric generation and load dispatch.
C311.5	Interpret the computer-controlled power system that employs PMU and SCADA.
Course Code & Title : EE3017 Embedded C programming	
Year / Semester : III Year/VI Semester	
Course Index : C312	
C312.1	Interpret the basic concepts of embedded C programming.
C312.2	Interpret the software and hardware architecture for embedded system.
C312.3	Evolve the C programs for 8051 microcontroller.
C312.4	Evolve the serial port and interrupt C programs for 8051 microcontroller.
C312.5	Evaluate the performance of the 8051 microcontroller programs used in various applications.
Course Code & Title : EE3020 Smart System Automation	
Year / Semester : III Year/VI Semester	
Course Index : C313	
C313.1	Interpret the performance of the components of the Smart system.
C313.2	Interpret the performance of the home automation system.
C313.3	Evaluate the performance of the smart energy systems.
C313.4	Evaluate the performance of the smart wearable devices.
C313.5	Evaluate the performance of the robots.
Course Code & Title : EE3035 Grid integrating Techniques and Challenges	
Year / Semester : III Year/VI Semester	
Course Index : C314	
C314.1	Interpret the current power scenario in India.
C314.2	Interpret the modeling of microgrid systems.
C314.3	Develop the converters for distributed generation systems.
C314.4	Interpret the various power quality measures in wind energy systems.
C314.5	Evaluate the performance of grid integrated renewable systems.

Course Code & Title : MX3089 Industrial Safety	
Year / Semester : III Year/VI Semester	
Course Index : C315	
C315.1	Demonstrate proficiency in applying fundamental safety principles.
C315.2	Execute practical demonstrations showcasing compliance with Statutory Regulations and standards.
C315.3	Perform safety activities within the working place
C315.4	Evaluate the effects of occupational exposures and their Remedies
C315.5	Apply and demonstrate proficiency in utilizing risk assessment techniques.
Course Code & Title : EE3611 Power System Laboratory	
Year / Semester : III Year/VI Semester	
Course Index : C316	
C316.1	Develop a program for determining the transmission line parameters and its electromagnetic transients.
C316.2	Build a simulation models to obtain the power flow solution using numerical iterative methods.
C316.3	Construct a simulation models to perform symmetrical, unsymmetrical fault analysis and analyze the stability of the power system and to solve economic dispatch problem
C316.4	Estimate the state of the power system and load frequency dynamics of interconnected power system.
C316.5	Develop a model for power system protection using relay, CT and PT.

Course Code & Title : EE3701 High Voltage Engineering Year / Semester : IV Year/VII Semester Course Index : C401	
C401.1	Interpret the causes of the over voltages and their protection methods in power systems.
C401.2	Select the relevant insulation materials based on the dielectric breakdown parameters.
C401.3	Describe the methods of generating and measuring high voltages and currents for the various applications using the relevant circuits.
C401.4	Perform the relevant 'Type' and 'Routine' tests for insulation breakdown
C401.5	Select the relevant high voltage/current equipment for various applications.
Course Code & Title : GE3791 Human Values and Ethics Year / Semester : IV Year/VII Semester Course Index : C402	
C402.1	Identify the importance of democratic, secular and scientific values in harmonious functioning of social life
C402.2	Explain the democratic and scientific values in both their personal and professional life
C402.3	Investigate the rational solutions to social problems.
C402.4	Express the ethical manner in society
C402.5	Discuss the critical thinking and the pursuit of truth.
Course Code & Title : GE3751 Principles of Management Year / Semester : IV Year/VII Semester Course Index : C403	
C403.1	Explain the evolution of management , different types of organization and current trends in management
C403.2	Discuss the planning process and explain the types of strategies in order to make rationalle decisions
C403.3	Elaborate different organizational structures and steps in an effective human resources management
C403.4	Assess different leadership styles, select appropriate styles for an organization and explain various theories of motivation
C403.5	Describe the control techniques and role of technology in Management
Course Code & Title : OCS352 IoT Concepts and Applications Year / Semester : IV Year/VII Semester Course Index : C404	
C404.1	Explain the concept of IoT
C404.2	Interpret the communication models and various protocols for IoT
C404.3	Apply data analytics and use cloud offerings related to IoT
C404.4	Design portable IoT using Arduino/Raspberry Pi /open platform
C404.5	Analyze applications of IoT in real time scenario

Course Code & Title : OHS351 English for Competitive Examinations	
Year / Semester : IV Year/VII Semester	
Course Index : C405	
C405.1	Communicate clearly in both written and oral forms using suitable vocabulary, while comprehending, understanding, analysing, and using language in competitive exams.
C405.2	Comprehend texts, and speak & write persuasively in academic and professional contexts mastering essential grammar rules and structures.
C405.3	Read, write, and speak effectively across various professional and social settings and critically analyze texts.
C405.4	Understand technical texts and enable them to effectively plan, structure, and develop various forms of written communication with a focus on coherence and style.
C405.5	Comprehend and respond to diverse spoken and written discourses in various accents while producing texts across different genres using several writing strategies.
Course Code & Title : CBM370 Wearable Devices	
Year / Semester : IV Year/VII Semester	
Course Index : C406	
C406.1	Describe the concepts of wearable system
C406.2	Explain the signal processing and energy harvestings in wearable device
C406.3	Recognize the concepts of wireless monitoring system and BAN in health care
C406.4	Illustrate the concept of smart textile and fabrication techniques
C406.5	Explain the various applications of wearable devices in healthcare system
Course Code & Title : EE3841 Project Work / Internship	
Year / Semester : IV Year/VII Semester	
Course Index : C407	
C407.1	The students will be able to apply the fundamental knowledge and skills, which are acquired within the technical area, to a given problem
C407.2	The students will be able to identify and summarize an appropriate list of literature review, analyze previous researchers' work and relate them to the project within given constraints, even with limited information, the students will be able to independently analyze and discuss complex inquiries/problems and handle larger problems on the advanced level within the technical area.
C407.3	The students will be able to design engineering solutions to complex problems in a systematic approach. The students will be able to identify and apply appropriate parameters, assumptions and design criteria in consideration of health and safety (example: the use of codes of practice), ethics, economics, environment, sustainability
C407.4	The students will be able to apply research and conduct experiments, as well as to analyze and interpret data that yield the results and answer important applicable research questions
C407.5	The students will be able to utilize technology tools for communication, collaboration, information management, and decision support.
C407.6	The students will be able to demonstrate the knowledge, skills and attitudes of a professional engineer.
C407.7	The students will be able to interact with team members in a professional manner, respecting differences, to ensure a collaborative project environment.
C407.8	The students will be able to demonstrate a strong working knowledge of ethics

	and professional responsibility.
C407.9	The students will be able to document and present one's own work, for a given target group, with strict requirements on structure, format, and language usage.
C407.10	The students will be able to present the project outlining the approach and expected results using good oral and written presentation skills
C407.11	The students will be able to demonstrate effective organizational leadership and change skills for managing projects and project teams
C407.12	The students will be able recognize the need for life-long learning by undergoing the project work.